experiences of space and time, it is not particularly surprising that a property relevant to the everyday fails to survive when pushed into the micro-realm. And since the arbitrary divisibility of space and time is one of their most familiar everyday properties, the inapplicability of this concept on ultrasmall scales gives another hint that there is something else lurking in the microdepths—something that might be called the bare-bones substrate of spacetime—the entity to which the familiar notion of spacetime alludes. We expect that this *ur*-ingredient, this most elemental spacetime stuff, does not allow dissection into ever smaller pieces because of the violent fluctuations that would ultimately be encountered, and hence is quite unlike the large-scale spacetime we directly experience. It seems likely, therefore, that the appearance of the fundamental spacetime constituents—whatever they may be—is altered significantly through the averaging process by which they yield the spacetime of common experience.

Thus, looking for familiar spacetime in the deepest laws of nature may be like trying to take in Beethoven's Ninth Symphony solely note by single note or one of Monet's haystack paintings solely brushstroke by single brushstroke. Like these masterworks of human expression, nature's spacetime whole may be so different from its parts that nothing resembling it exists at the most fundamental level.

Geometry in Translation

Another consideration, one physicists call *geometrical duality,* also suggests that spacetime may not be fundamental, but suggests it from a very different viewpoint. Its description is a little more technical than quantum averaging, so feel free to go into skim mode if at any point this section gets too heavy. But because many researchers consider this material to be among string theory's most emblematic features, it's worth trying to get the gist of the ideas.

In Chapter 13 we saw how the five supposedly distinct string theories are actually different translations of one and the same theory. Among other things, we emphasized that this is a powerful realization because, when translated, supremely difficult questions sometimes become far simpler to answer. But there is a feature of the translation dictionary